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3-2: Two Numbers, Two Jobs

TEKS P.5.J: solve polynomial equations by applying a variety of techniques. Today: factor $x^2 + bx + c$ and solve by factoring.

Warm-up: Number detective

Find two integers with the given product and the given sum. The first row is done for you.

Product	Sum	The two numbers
15	8	3 and 5
12	7	
12	−7	
−12	1	
−12	−4	

Part A: Multiply, then look

Multiply each product (FOIL or a box). Then, using the two numbers p and q inside the parentheses, fill in $p + q$ and $p \cdot q$.

Product $(x + p)(x + q)$	Expanded: $x^2 + bx + c$	$p + q$	$p \cdot q$
$(x + 3)(x + 4)$			
$(x - 3)(x - 4)$			
$(x + 6)(x - 2)$			
$(x - 6)(x + 2)$			

What do you notice about how $p + q$ and $p \cdot q$ show up in the expanded form? Look at the last two rows: what changed when the signs of p and q flipped?

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Part B: Sign detective

Use the warm-up and Part A. For each sign of b and c in $x^2 + bx + c$, what are the signs of the two numbers? If they are different, which one is bigger (in size)?

b	c	Signs of the two numbers	An example from today
+	+		
−	+		
+	−		
−	−		

Practice

Factor each trinomial. List the factor pairs of c, circle the pair that adds to b, and check by multiplying. Tags: **B** = Basic (leading coefficient 1), **I** = Intermediate (leading coefficient greater than 1).

1. $x^2 + 8x + 15$ **B**

2. $x^2 - 9x + 20$ **B**

3. $x^2 + 2x - 15$ **B**

4. $x^2 - 3x - 10$ **B**

5. $x^2 + 5x - 36$ **B** (many pairs: be systematic)

6. $x^2 - 10x + 25$ **B**

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Practice (continued)

Problems 7, 8, and 12 start with a leading coefficient greater than 1. (Hint: what did we do first in 3-1?) For 9, factor or show why you cannot.

7. Factor completely: $2x^2 + 14x + 24$ I

8. Factor completely: $4x^2 - 4x - 24$ I

9. $x^2 + 3x + 5$ B (Factor it, or show why you cannot.)

10. Solve: $x^2 - 7x + 10 = 0$ B

11. Solve: $x^2 + 4x = 21$ B

12. Solve: $3x^2 - 15x + 12 = 0$ I

13. B A rectangular patio has area $\mathbf{x^2 + 11x + 28}$ square feet, where x is in feet.

(a) Factor to write the length and the width as expressions in x.

(b) The patio's area is actually 70 square feet. Write an equation equal to zero, solve it, and give the real length and width. Which solution makes sense, and why?

Math Time Machine: Babylonian scribes (about 2000 to 1600 BCE) solved problems like these on clay tablets with no algebra symbols. In the 1590s, François Viète found the sum-and-product link between roots and coefficients. Thomas Harriot (published in 1631) was among the first to put zero on one side and factor, and Descartes (1637) helped give us x for unknowns and a, b, c for knowns.